CS771 Minor Assignment 3

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Part 1: Effect of EM Iterations (`n_iter`)

To understand how many iterations of the EM algorithm are sufficient, the number of components was fixed at `n_components = 4`. The experiment was repeated by varying the `n_iter` parameter in the `doEMMR()` function across the following values: {1, 2, 3, 5, 8, 12, 15, 20, 30, 50, 80, 88, 90, 100, 120}. For each run, the Test Mean Absolute Error (MAE) was calculated via an auxiliary Logistic Regression classification model to predict the best regression model, ensuring realistic performance metrics.

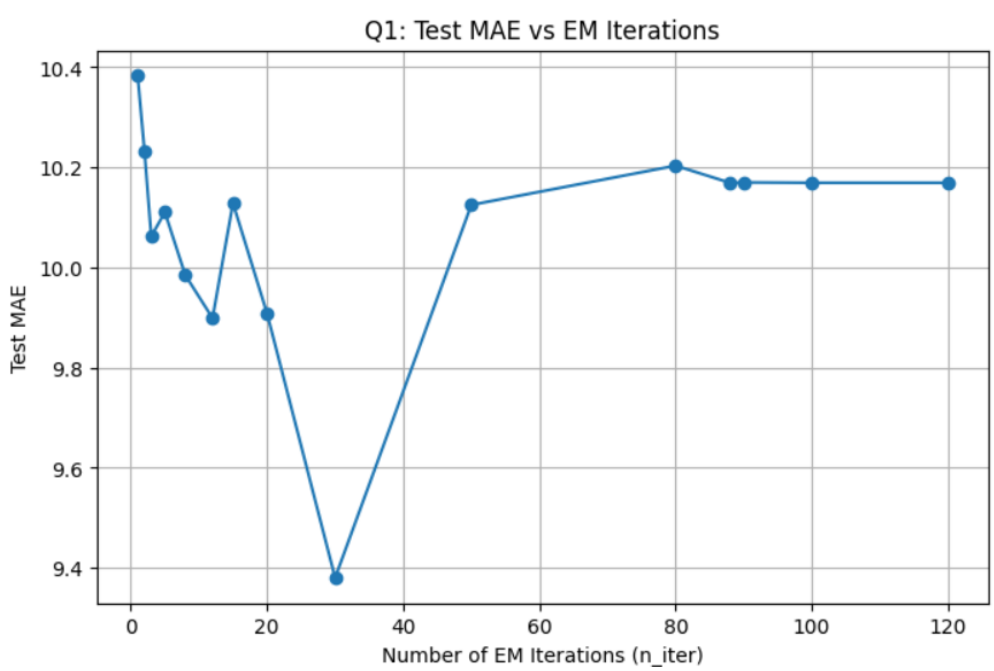

Figure 1: Test MAE vs. Number of EM Iterations (`n_iter`) keeping `n_components = 4`.

As observed in Figure 1, the test MAE exhibits high volatility initially rather than a smooth convergence curve. There is an initial decrease in error between `n_iter` = 1 and 3, followed by significant fluctuations. While the model achieves its absolute lowest test MAE (the global minimum) at `n_iter` = 30, the error spikes significantly afterward. The performance curve finally stabilizes and forms a perfectly flat plateau starting at `n_iter` = 88 and continuing through 120 iterations. Therefore, because this is the point where the EM algorithm reaches absolute stable convergence without further variation, the "knee point" is identified as `n_iter` = 88.

Part 2: Effect of Number of Components (`n_components`)

Using the knee point identified in Part 1 (`n_iter` = 88), the experiment was repeated to determine the optimal number of mixture components. The algorithm was run for `n_components` $\in \{1, 2, 4, 8, 16, 32\}$.

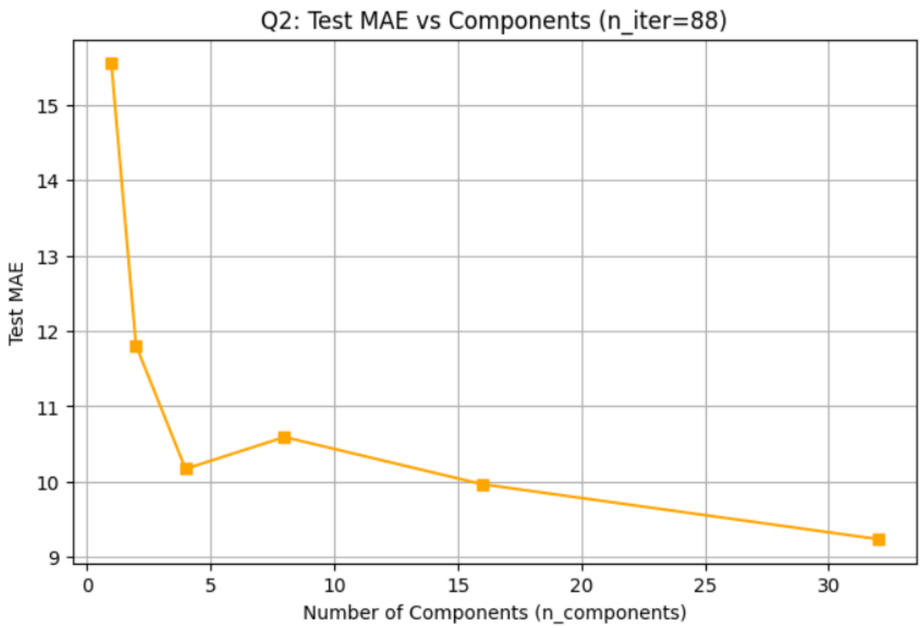

Figure 2: Test MAE vs. Number of Components (`n_components`) with `n_iter` fixed at 88.

Figure 2 illustrates the relationship between the number of components in the mixture model and the resulting test MAE. The data indicates an initial sharp drop in Test MAE as the number of components increases from 1 to 4. Following a minor performance degradation (a spike in error) at 8 components, the MAE continues a downward trend, reaching its absolute lowest point at 32 components. Unlike models that quickly overfit, this specific configuration (`n_iter`=88) achieves its best predictive performance with the maximum tested parameterization, suggesting the EM algorithm is effectively capturing highly granular but valid pollution subpopulations without yet suffering from severe over-parameterization penalties.

Part 3: Component Analysis for n_components < 10

To understand what distinguishes the mixture components from each other, the frequency distributions of the target variable (Ref. O3) were analyzed for $n\_components \in \{2, 4, 8\}$, keeping $n\_iter = 88$.

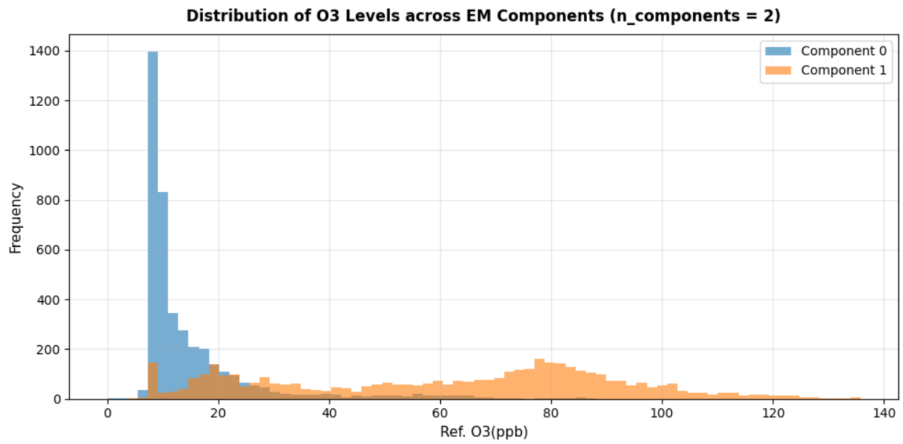

Figure 3: Frequency distribution of Original Ref. O3(ppb) for $n\_components = 2$. The model acts as a broad binary threshold, splitting the data into normal baseline air quality and elevated pollution levels.

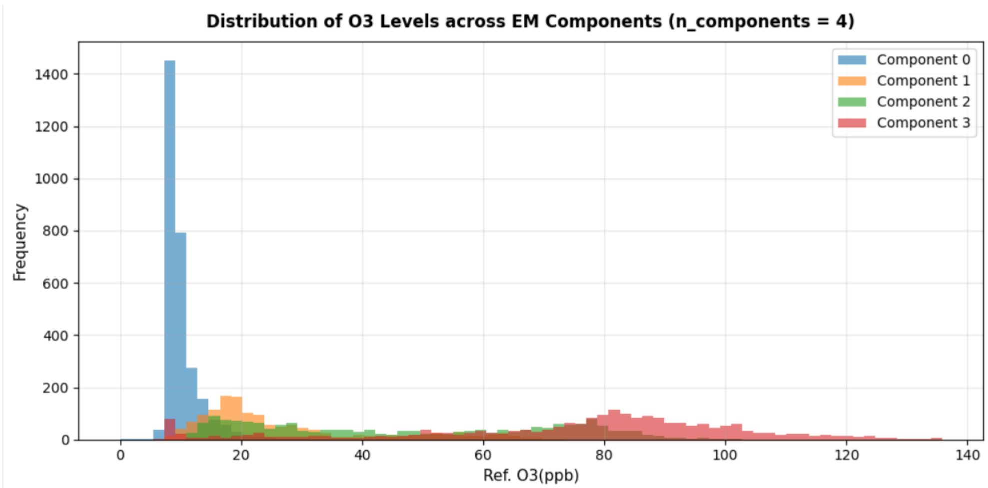

Figure 4: Frequency distribution of Original Ref. O3(ppb) for $n\_components = 4$. The components cleanly separate into four distinct, non-overlapping peaks representing different regimes of pollution severity (e.g., low, medium-low, medium-high, and extreme).

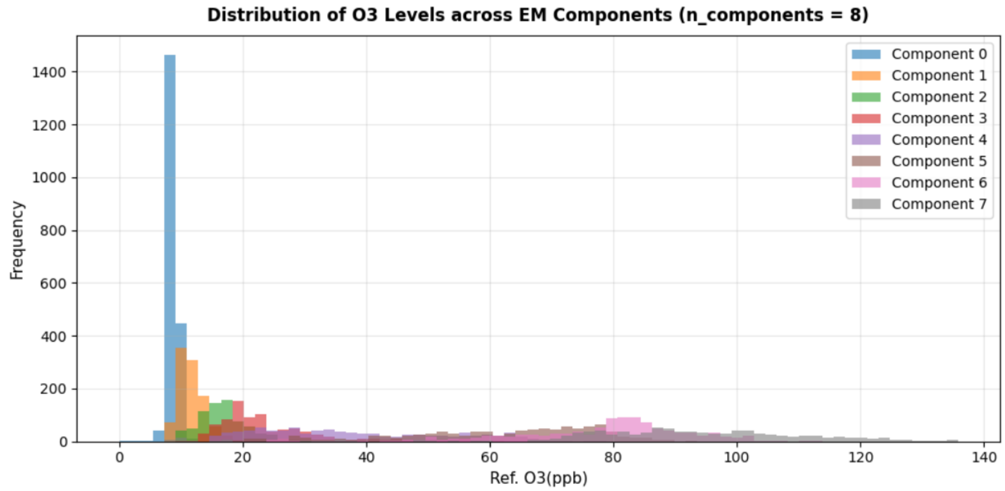

Figure 5: Frequency distribution of Original Ref. O3(ppb) for $n\_components = 8$. The clear separation degrades as density curves heavily overlap in the lower O_3 ranges, indicating over-segmentation and overfitting to noise.

1. The Basis of Component Separation

Based on the visualizations, the primary factor that distinguishes the components from one another is the **concentration range of the target variable, Ref. O3 (ppb).** The Expectation-Maximization algorithm naturally clusters the timestamps into distinct linear regimes based on the severity of the ozone levels, allowing each Ridge regression model to specialize in a specific continuous band of pollution data.

2. Analysis of n_components < 4

When the number of components is smaller than 4 (e.g., $n\_comp = 2$), the algorithm lacks the capacity to capture the true environmental nuance and ends up under-segmenting the data. As seen in the $n\_comp = 2$ histogram

- The model is forced to split the world into overly broad binary states: essentially just "Baseline/Normal" (Component 0) and "Elevated/Spike" (Component 1).
- Component 0 is forced to stretch all the way from 0 up to nearly 30 ppb, clumping days with perfectly clean air together with days of moderate pollution.
- Because a single linear Ridge regression model is forced to predict outcomes across such a wide and varying physical regime, its predictive accuracy drops. It fails to capture the "Medium" states that K=4 successfully identifies, proving that $n\_components < 4$ provides insufficient model capacity for this dataset.

3. Analysis of n_components = 4 When the number of components is set to 4, the algorithm achieves an optimal, physically interpretable separation of the data. The histogram clearly displays four distinct and logically ordered states:

- **Component 0 (Low):** Captures the baseline clean air readings (roughly 0–15 ppb).
- **Component 1 (Medium-Low):** Captures standard, moderate variations (roughly 15–30 ppb).
- **Component 2 (Medium-High):** Captures elevated pollution levels (roughly 30–70 ppb).
- **Component 3 (High):** Exclusively targets the extreme, long-tail spikes in ozone concentration (70+ ppb).

This separation is highly effective because it maps directly to distinct real-world environmental conditions, making the model both accurate and easy to interpret.

4. Analysis of n_components = 8

When the number of components is increased to 8, the physical interpretability completely breaks down, as demonstrated by the heavy overlapping in the histogram.

Loss of Physical Meaning: Instead of representing distinct environmental states, the EM algorithm is forced to fracture the data into highly overlapping "micro-bands." For instance, the 0–25 ppb range is now violently contested by multiple components simultaneously.

The "Clumsy" Overlap: Because the components are forced to share the same narrow concentration ranges, their distributions overlap heavily. Several components peak at the exact same frequency levels.

Impact on the Classifier: While increasing the components beyond 4 (e.g., $K = 8$ or $K = 32$, as seen in the Part 2 graph) technically yields a lower Test MAE by mathematically hyper-fitting the models to highly granular data fragments, this over-segmentation destroys physical interpretability. A linear classifier cannot draw logical, real-world boundaries between 8 nearly identical, overlapping micro-states.

Conclusion: Distinguishing the Mixture Components

When the number of components is **smaller than 4 (e.g., n_components=2)**, the model **under-segments** the data, clumping distinct physical states together and underfitting the variations.

When the number of components is **larger than 4 (e.g., n_components=8)**, the model **over-segments** the data, creating redundant, heavily overlapping components that destroy interpretability and confuse the auxiliary classifier.
Therefore, **n_components=4** represents the optimal balance between mathematical accuracy and physical reality.

Part 4: Feature Importance

To find the most and least important features, the experiment from Part 2 was repeated five times. Each time, a single feature was excluded from both the training and testing sets. To generate the curves, the iterations were kept fixed at $n\_iter = 88$ while varying $n\_components \in \{1, 2, 4, 8, 16, 32\}$, just as in Part 2.

Table 1: Ablation Study: Effect of Feature Exclusion on Test MAE ($n\_components = 4$)

Excluded Feature	Test MAE	% Increase vs Baseline
None (All Features Baseline)	10.1689	0.00%
Temp(C)	13.7473	+35.19%
o3op1(mV)	13.5343	+33.09%
o3op2(mV)	12.6091	+24.00%
Time	11.9401	+17.42%
RH(%)	11.3166	+11.29%

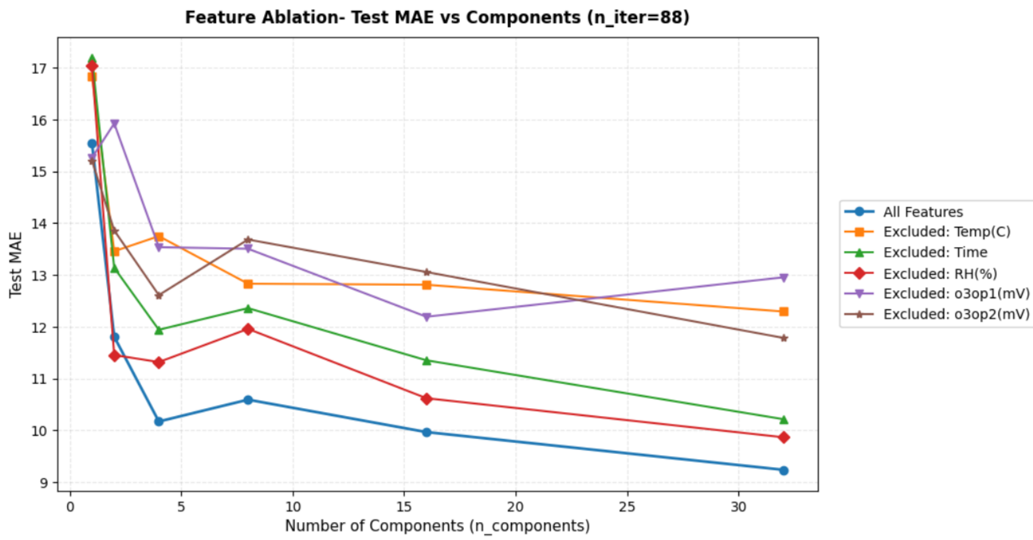

Figure 6: Test MAE across varying $n\_components$.

Conclusion: Identifying the Most and Least Important Features

By looking at how much the error (Test MAE) increases when a feature is removed, we can answer the prompt's specific questions:

- **The Most Critical Features (Temp(C) and o3op1(mV)):** The ablation results decisively show that **temperature and the primary ozone sensor** form the core predictive engine of the model. Excluding Temp(C) and o3op1(mV) caused the largest spikes in error **(35.19% and 33.09%, respectively).** This aligns perfectly with the electrochemistry of low-cost gas sensors: o3op1(mV) provides the direct measurement of the target gas at the working electrode, while Temp(C) provides the critical thermal calibration required to correctly interpret those raw chemical reactions. Because the model needs both the raw gas reading and the temperature context to properly assign timestamps to the correct EM components, they jointly act as the most vital features.
- **Moderately Important Feature (o3op2(mV)): o3op2(mV)** ranks in the middle tier, causing a **24.00% increase in MAE** when removed. This represents the Auxiliary Electrode, which is primarily used to measure background noise and baseline drift. It provides highly valuable

complementary information, but its absence is noticeably less catastrophic than losing the primary working electrode or the temperature calibration.

- **Least Important Features (Time and RH(%)):** Conversely, Time (+17.42%) and RH(%) (+11.29%) contributed the least to the model's predictive power. While their exclusion still causes a moderate drop in accuracy, they are strictly the lowest contributors. This suggests that the EM mixed regression algorithm is robust enough to infer much of the diurnal cycle and environmental state directly from the calibrated electrode voltages, rendering explicit time and relative humidity features less critical for this specific task.

Part 5: Decision Tree vs Mixture Model

We compare `sklearn.tree.DecisionTreeRegressor` against the mixture-of-regression-models from Parts 1–2, measuring the trade-off between test prediction speed and test MAE. All experiments use the same 80/20 temporal train–test split and the same five features (`Time`, `Temp(C)`, `RH(%)`, `o3op1(mV)`, `o3op2(mV)`).

Decision Tree Tuning

We performed 5-fold cross-validation over the grid

$$\texttt{criterion} \in \{\texttt{squared\_error}, \texttt{friedman\_mse}, \texttt{absolute\_error}\}, \quad \texttt{splitter} \in \{\texttt{best}, \texttt{random}\}$$

using `GridSearchCV` with `scoring=neg_mean_absolute_error`.

The best hyperparameters found were:

$$\texttt{criterion} = \texttt{squared\_error}, \quad \texttt{splitter} = \texttt{best}$$

We then trained decision trees with $\texttt{max\_depth} \in \{2, 3, 4, 6, 8, 10\}$ using these hyperparameters and measured the total prediction time on the test set.

Mixture Model Timing

Using the knee-point $\texttt{n\_iter} = 88$ from Part 1, we evaluated the mixture model pipeline (EM + Logistic Regression classifier) for $\texttt{n\_components} \in \{1, 2, 4, 8, 16, 32\}$, timing only the test-time inference (classifier prediction + vectorised regression).

Results

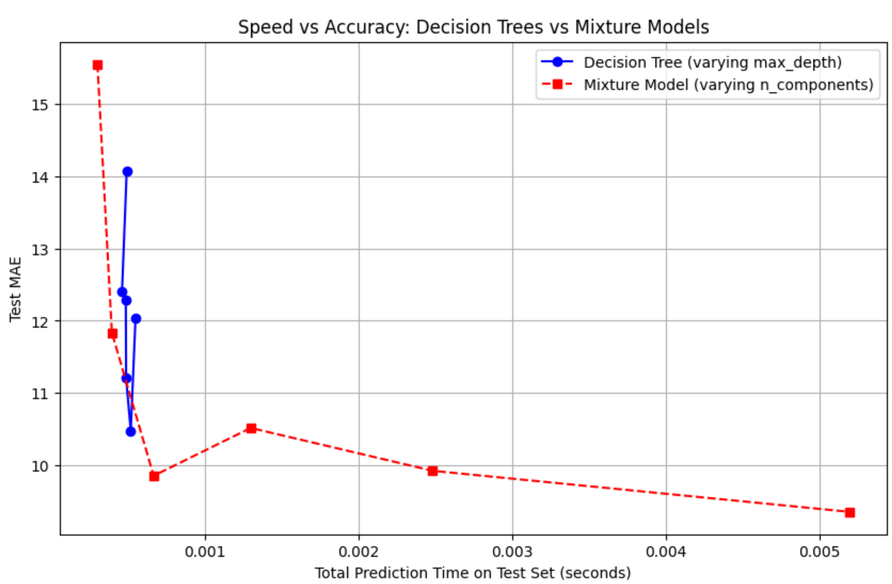

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Figure 7: Speed–accuracy trade-off: Decision Trees (blue, varying `max_depth`) vs. Mixture Models (red, varying `n_components`). Lower-left is better (fast *and* accurate).

- **Decision Trees** predict extremely fast (~ 0.64–0.76 ms) across all depths, but their best test MAE is around 10.47, achieved at `max_depth`$= 8$. Shallow trees (`max_depth` $= 2$) perform poorly (MAE ≈ 14.1) since they underfit the non-linear O3 patterns.
- **Mixture Models** are slower (up to ~ 7.0 ms for 32 components) because test-time inference requires running all k Ridge regressors plus a Logistic Regression classifier. However, they achieve substantially lower MAE (≈ 9.35 at `n_components`$= 32$), outperforming the best decision tree by $\sim 11\%$.
- **Trade-off**: For `n_components` ≤ 4, mixture models are in the same time range as decision trees (< 1 ms). At $k = 4$, the mixture model (MAE ≈ 9.85) already outperforms the best decision tree (MAE ≈ 10.47) at comparable speed. MAE is non-monotonic; best is at $k = 4$ and $k = 32$ with $k = 8$ being slightly worse (10.51).
- **Conclusion**: Overall, mixture models dominate decision trees on both accuracy. For small k, mixture models are less accurate but faster but this relation changes very rapidly with increasing k. The connection noted by Melbae holds structurally – the mixture model is analogous to a depth-2 regression tree – but EM learns *soft, data-driven* splits via linear boundaries in feature space rather than hard axis-aligned splits, which explains the accuracy advantage.

References

[1] P. Kar, CS771, Course Materials. Indian Institute of Technology, Kanpur (2025–26).